



WAZUH + SURICATA



Network Intrusion Detection Setup Guide

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Objective

To explore how an Endpoint Detection and Response (EDR) system can effectively detect suspicious network activity by integrating **Suricata** (an Intrusion Detection System) and visualizing real-time alerts through the **Wazuh** security platform.

What is Suspicious Network Traffic?

Suspicious network traffic is unusual or unexpected activity that may indicate malicious behavior.

Examples:

- Port scanning (e.g., Nmap)
- Connections to known malicious IPs (C2 servers)
- FTP uploads on non-standard ports
- HTTP requests to rare or suspicious domains

How Are IDSs Helpful?

Intrusion Detection Systems (IDS) enhance security by:

-  Monitoring network traffic in real time
-  Alerting on known attack signatures
-  Detecting reconnaissance, malware activity, and lateral movement
-  Providing early threat visibility to SOC teams

How Does IDS Work?

-  **Packet Capture** – Inspects incoming and outgoing network packets
-  **Rule Matching** – Compares traffic against predefined attack signatures

- 🚨 **Alerting** – Generates alerts when suspicious activity is detected
- 🔗 **Integration** – Forwards alerts to SIEMs like **Wazuh** for analysis and visualization

🦆 What is Suricata?

Suricata is an open-source, high-performance **Network IDS/IPS** and traffic monitoring engine.

🔧 Key Features:

- 🔍 Deep packet inspection
- 🧠 Multi-threaded performance
- 🌐 Protocol detection (HTTP, TLS, DNS, etc.)
- 📄 JSON logging for easy integration
- 🛡️ Community-driven rule sets for threat detection
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💻 Lab Setup

Component	Description
🛡️ Wazuh Server	Ubuntu Server running Wazuh Manager and Dashboard
🔍 Wazuh Agent	Ubuntu system with Suricata IDS installed
⚔️ Attacker Machine	Kali Linux used to simulate suspicious or malicious traffic

Step 1: Installing Suricata and Rules

1. Install Suricata on the Ubuntu endpoint. We tested this process with version 6.0.8 and it can take some time:

```
sudo add-apt-repository ppa:oisf/suricata-stable
```

```
sudo apt-get update
```

```
sudo apt-get install suricata -y
```

```
root@kali:~/home/kali# sudo add-apt-repository ppa:oisf/suricata-stable
Add repository 'ppa:oisf/suricata-stable'
root@kali:~/home/kali# sudo apt-get update
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [51.4 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [126 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [327 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [197 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [211 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [10.6 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [26.4 kB]
Fetched 74.0 MB in 1min 18s (1,059 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libfuse3-3
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libbftd1 libbft2 libhyperscan5 libnetfilter-log1 librtb-bus-pci25 librtb-bus-vdev25 librtb-eal25 librtb-ethdev25 librtb-hash25 librtb-ip-frag25 librtb-kvargs25 librtb-log25 librtb-mbuf25 librtb-mempool25 librtb-meter25
  librtb-net-bond25 librtb-net25 librtb-pci25 librtb-rcu25 librtb-ring25 librtb-sched25 librtb-telemetry25 libxdp1 oinkmaster snort-rules-default sse4.2-support suricata suricata-update
Suggested packages:
  snort | snort-psql | snort-mysql libtool-minimal4
The following NEW packages will be installed:
  libbftd1 libbft2 libhyperscan5 libnetfilter-log1 librtb-bus-pci25 librtb-bus-vdev25 librtb-eal25 librtb-ethdev25 librtb-hash25 librtb-ip-frag25 librtb-kvargs25 librtb-log25 librtb-mbuf25 librtb-mempool25 librtb-meter25
  librtb-net-bond25 librtb-net25 librtb-pci25 librtb-rcu25 librtb-ring25 librtb-sched25 librtb-telemetry25 libxdp1 oinkmaster snort-rules-default sse4.2-support suricata suricata-update
0 upgraded, 28 newly installed, 0 to remove and 65 not upgraded.
Need to get 6,971 kB of archives.
After this operation, 32.0 MB of additional disk space will be used.
Get:1 http://kali.download/kali kali-rolling/main amd64 sse4.2-support amd64 27 [3,695 B]
Get:2 http://kali.download/kali kali-rolling/main amd64 libbftd1 amd64 1.0.5.50-1 [72.9 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 libhyperscan5 amd64 5.4.2-3 [2,695 kB]
Get:4 http://mirror.ourhost.az/kali kali-rolling/main amd64 librtb-kvargs25 amd64 24.11.2-2 [17.8 kB]
Get:5 http://mirror.ourhost.az/kali kali-rolling/main amd64 librtb-telemetry25 amd64 24.11.2-2 [26.7 kB]
Get:6 http://mirror.kku.ac.th/kali kali-rolling/main amd64 snort-rules-default all 3.1.02.0-0kali1 [220 kB]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 libnetfilter-log1 amd64 1.0.2-4+b1 [13.3 kB]
Get:8 http://http.kali.org/kali kali-rolling/main amd64 libbftd1 amd64 1.7.2-2+b1 [20.0 kB]
Get:9 http://kali.download/kali kali-rolling/main amd64 librtb-log25 amd64 24.11.2-2 [23.7 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 librtb-eal25 amd64 24.11.2-2 [159 kB]
```

Step 2: Download and extract the Emerging Threats Suricata ruleset:

```
cd /tmp/ && curl -LO
```

<https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz>

```
sudo tar -xvzf emerging.rules.tar.gz && sudo mkdir /etc/suricata/rules && sudo mv
rules/*.rules /etc/suricata/rules/
```

```
sudo chmod 640 /etc/suricata/rules/*.rules
```

```

# cd /tmp/ && curl -LO https://rules.emergingthreats.net/open/suricata-6.0.8/emerging.rules.tar.gz
sudo tar -xvzf emerging.rules.tar.gz && sudo mkdir /etc/suricata/rules && sudo mv rules/*.rules /etc/suricata/rules/
sudo chmod 640 /etc/suricata/rules/*.rules
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed                               Day29: EDR Basics: Detecting Suspicious Network Traffic.md
100 4871k  100 4871k    0     0 84143      0  0:00:59  0:00:59 --:--:--  966k
rules/
rules/BSD-License.txt
rules/LICENSE
rules/botcc.portgrouped.rules
rules/botcc.rules
rules/ciarmy.rules
rules/classification.config
rules/compromised-ips.txt
rules/compromised.rules
rules/drop.rules
rules/dshield.rules
rules/emerging-activex.rules
rules/emerging-adware_pup.rules
rules/emerging-attack_response.rules
rules/emerging-chat.rules
rules/emerging-coinminer.rules
rules/emerging-current_events.rules
rules/emerging-deleted.rules
rules/emerging-dns.rules
rules/emerging-dos.rules
rules/emerging-exploit.rules
rules/emerging-exploit_kit.rules
rules/emerging-ftp.rules
rules/emerging-games.rules
rules/emerging-hunting.rules
rules/emerging-icmp.rules
rules/emerging-icmp_info.rules
rules/emerging-icmp.rules
rules/emerging-inappropriate.rules
rules/emerging-info.rules
rules/emerging-ja3.rules
rules/emerging-malware.rules
rules/emerging-misc.rules
rules/emerging-mobile_malware.rules
rules/emerging-netbios.rules
rules/emerging-p2p.rules
rules/emerging-phishing.rules
rules/emerging-policy.rules
rules/emerging-pop3.rules
rules/emerging-retired.rules
rules/emerging-rpc.rules
rules/emerging-scada.rules
rules/emerging-scan.rules
rules/emerging-shellcode.rules
rules/emerging-smtp.rules
rules/emerging-smnp.rules
rules/emerging-sql.rules
rules/emerging-tlsnet.rules
rules/emerging-tftp.rules
rules/emerging-user_agents.rules
rules/emerging-voip.rules
rules/emerging-web_client.rules
rules/emerging-web_server.rules
rules/emerging-web_specific_apps.rules

```

Task: Detecting Port Scanning using Suricata + Wazuh

1. Install Suricata on the Ubuntu endpoint. We tested this process with version 6.0.8 and
2. Download and extract the Emerging Threats Suricata ruleset:
3. Restart the Suricata service:
4. Add the following configuration to the /var/ossec/etc/ossec.conf file of the Wazuh agent. The

```

ossec.conf
<localrule>
  <sig> <file> /tmp/10g_for_kali
  <localrule> /var/log/suricata/eve - /tmp/location
</localrule>
</ossec.conf>

```

Step 3: Simulate Attack using Kali Linux

Step 3: Restart the Suricata service:

```
sudo systemctl restart suricata
```

Step 4: Configure Suricata

1. Open the config file:

```
sudo nano /etc/suricata/suricata.yaml
```

Update network variables:

HOME_NET: "[192.168.201.24]"

EXTERNAL_NET: "!\$HOME_NET"

```
YAML 1.1
# Suricata configuration file. In addition to the comments describing all
# options in this file, full documentation can be found at:
# https://docs.suricata.io/en/latest/configuration/suricata-yaml.html
# This configuration file generated by Suricata 7.0.10.
suricata-version: "7.0"

##
## Step 1: Inform Suricata about your network
##

vars:
  # more specific is better for alert accuracy and performance in the example below
  address-groups:
    HOME_NET: "[192.168.201.24]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    EXTERNAL_NET: "!$HOME_NET"
    #EXTERNAL_NET: "any"

    HTTP_SERVERS: "$HOME_NET"
    SMTP_SERVERS: "$HOME_NET"
    SQL_SERVERS: "$HOME_NET"
    DNS_SERVERS: "$HOME_NET"
    TELNET_SERVERS: "$HOME_NET"
    AIM_SERVERS: "$EXTERNAL_NET"
    DC_SERVERS: "$HOME_NET"
    DNP3_SERVER: "$HOME_NET"
    DNP3_CLIENT: "$HOME_NET"
    MODBUS_CLIENT: "$HOME_NET"
    MODBUS_SERVER: "$HOME_NET"
    ENIP_CLIENT: "$HOME_NET"
    ENIP_SERVER: "$HOME_NET"

  port-groups:
    #EXTERNAL_NET: "any"
    HTTP_PORTS: "80"
    SHELLCODE_PORTS: "180"
    ORACLE_PORTS: "1521"
    SSH_PORTS: "22"
    DNP3_PORTS: "20000"
    MODBUS_PORTS: "502"
    FILE_DATA_PORTS: "[$HTTP_PORTS,110,143]"
    FTP_PORTS: "21"
    GENEVE_PORTS: "6081"
    VXLAN_PORTS: "4789"
    TEREDO_PORTS: "3544"

##
## Step 2: Select outputs to enable
##

# Suricata IDS and Wazuh
```

Step 5:Set network interface (replace wlan0with yours)

Af-packet:

- interface: wlan0

```
# Define your logging outputs. If none are defined, or they are all
# disabled you will get the default: console output.
outputs:
- console:
    enabled: yes
    # type: json
- file:
    enabled: yes
    level: info
    filename: suricata.log
    # format: "[%i - %m] %z %d: %S: %M"
    # type: json
- syslog:
    enabled: no
    facility: local5
    format: "[%i] <%d> -- "
    # type: json

Step 3: Configure common capture settings

See "Advanced Capture Options" below for more options, including Netmap
and PF_RING.

Linux high speed capture support
-packet:
- interface: wlan0
  # Number of receive threads. "auto" uses the number of cores
  #threads: auto
  # Default clusterid. AF_PACKET will load balance packets based on flow.
  cluster-id: 99
  # Default AF_PACKET cluster type. AF_PACKET can load balance per flow or per hash.
  # This is only supported for Linux kernel > 3.1
  # possible value are:
  # * cluster_flow: all packets of a given flow are sent to the same socket
  # * cluster_cpu: all packets treated in kernel by a CPU are sent to the same socket
  # * cluster_qm: all packets linked by network card to a RSS queue are sent to the same
  # socket. Requires at least Linux 3.14.
  # * cluster_ebpf: eBPF file load balancing. See doc/userguide/capture-hardware/ebpf-xdp.rst for
  # more info.
  # Recommended modes are cluster_flow on most boxes and cluster_cpu or cluster_qm on system
  # with capture card using RSS (requires cpu affinity tuning and system IRQ tuning)
  # cluster_rollover has been deprecated; if used, it'll be replaced with cluster_flow.
  cluster-type: cluster_flow
  # In some fragmentation cases, the hash can not be computed. If "defrag" is set
  # to yes, the kernel will do the needed defragmentation before sending the packets.
  defrag: yes
  # To use the ring feature of AF_PACKET, set 'use-mmap' to yes
  #use-mmap: yes
  # Lock memory map to avoid it being swapped. Be careful that over
  # subscribing could lock your system
  #mmap-locked: yes
  # Use tpacket v3 capture mode, only active if use-mmap is true
  # Don't use it in IPS or TAP mode as it causes severe latency
```

Step: 6

Restart Suricata:

sudo systemctl restart suricata

sudo systemctl enable suricata

Step 7: Already Installed Wazuh Agent on kali-NIDS

Configure Wazuh Agent:

sudo nano /var/ossec/etc/ossec.conf

Set manager IP:

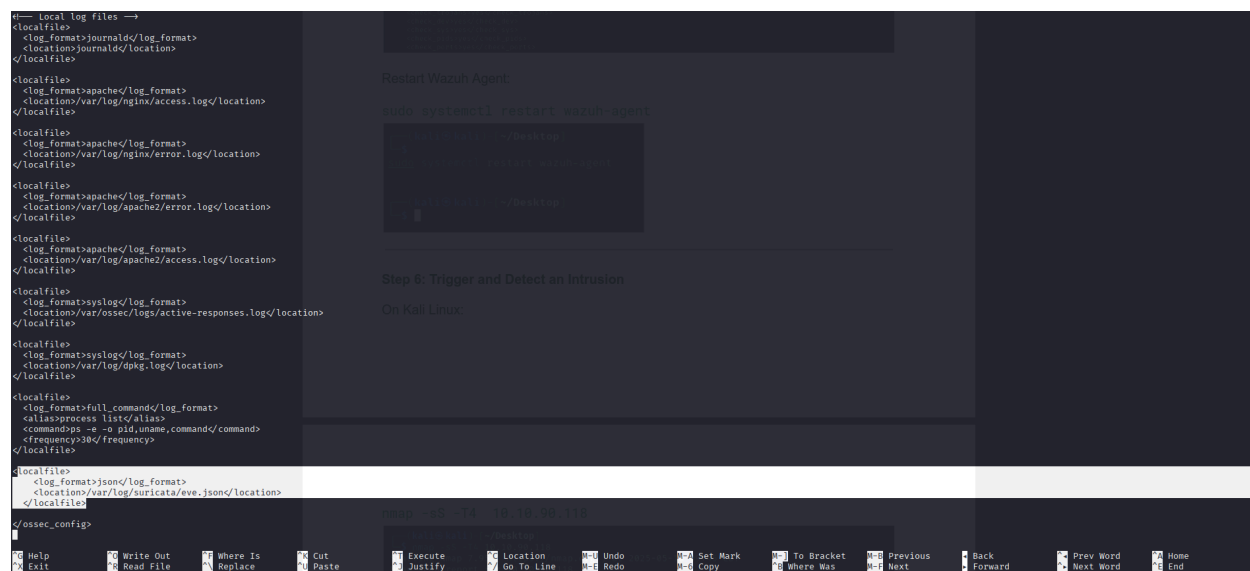
<client>

<server>

<address>10.10.90.117</address>

</server>

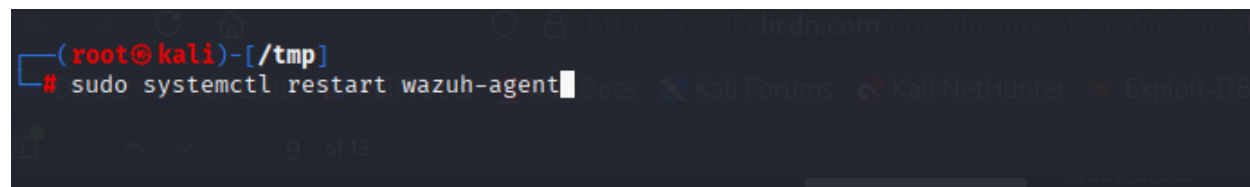
</client>



Step 8:

Restart Wazuh Agent:

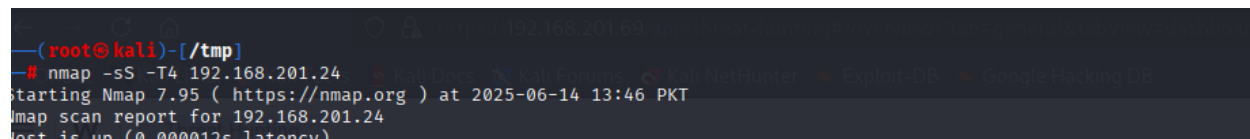
`sudo systemctl restart wazuh-agent`



Step 9: Trigger and Detect an Intrusion

On Kali Linux:

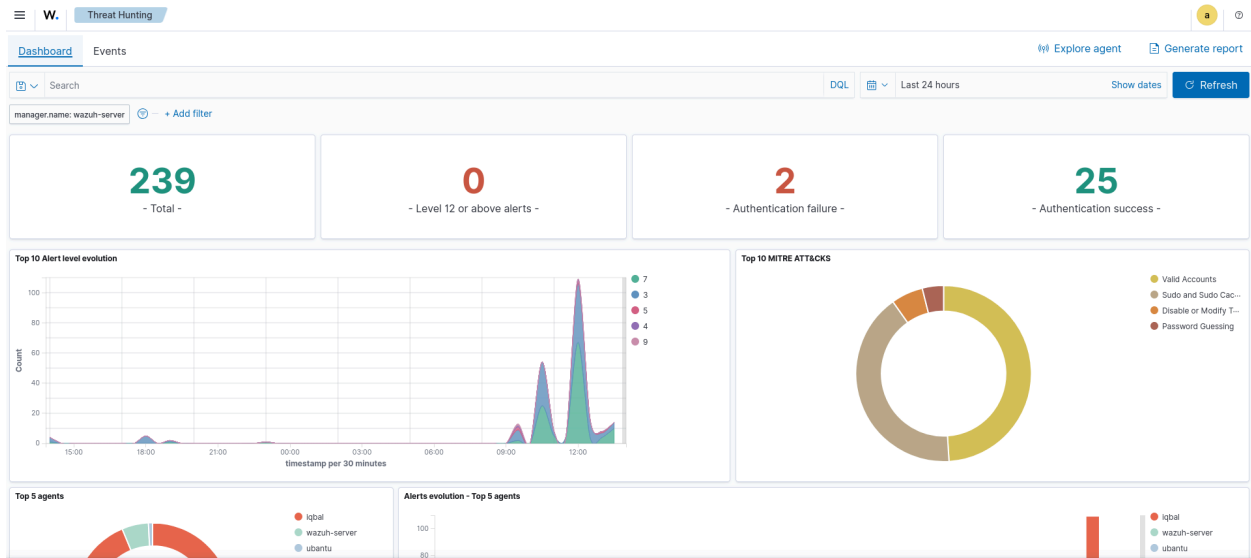
`nmap -sS -T4 192.168.201.24`



Step 10:

Then, log into Wazuh Dashboard > Modules > Security Events or NIDS

Look for Suricata alerts (e.g., Nmap scan detection).



Step 11: Investigate and Respond

The screenshot shows the Wazuh Dashboard interface for the Events module. At the top, there are navigation tabs for 'Dashboard' and 'Events'. Below the navigation bar, there are four summary cards: '239 hits', '497 columns hidden', 'Density', and '1 fields sorted'. The main content area features a bar chart showing alert counts over time. Below the chart, there is a table of alerts with columns: timestamp, agent.name, rule.description, rule.level, and rule.id.

timestamp	agent.name	rule.description	rule.level	rule.id
Jun 14, 2025 @ 13:45:34.996	iqbal	Listened ports status (netstat) changed (new port op...	7	533
Jun 14, 2025 @ 13:44:46.843	iqbal	Suricata: Alert - SURICATA STREAM excessive retrans...	3	86601
Jun 14, 2025 @ 13:39:38.489	iqbal	PAM: Login session closed.	3	5502
Jun 14, 2025 @ 13:39:33.274	iqbal	Host-based anomaly detection event (rootcheck).	7	510
Jun 14, 2025 @ 13:39:33.246	iqbal	Host-based anomaly detection event (rootcheck).	7	510
Jun 14, 2025 @ 13:39:33.104	iqbal	Host-based anomaly detection event (rootcheck).	7	510
Jun 14, 2025 @ 13:39:33.098	iqbal	Host-based anomaly detection event (rootcheck).	7	510
Jun 14, 2025 @ 13:39:33.092	iqbal	Host-based anomaly detection event (rootcheck).	7	510
Jun 14, 2025 @ 13:39:33.086	iqbal	Host-based anomaly detection event (rootcheck).	7	510

- Use Suricata logs for forensic review.

Document Details

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[Table](#)

JSON

t _index	wazuh-alerts-4.x-2025.06.14
t agent.id	001
t agent.ip	192.168.100.235
t agent.name	Iqbal
data.alert.action	allowed
data.alert.category	Generic Protocol Command Decode
data.alert.gid	1
data.alert.rev	1
data.alert.severity	3
data.alert.signature	SURICATA STREAM excessive retransmissions
data.alert.signature_id	2210054
data.app_proto	tls
data.dest_ip	2a00:1450:4019:0003:0000:0000:0000:0006
data.dest_port	443
data.direction	to_server
data.event_type	alert
data.flow.bytes_toclient	8232
data.flow.bytes_toserver	22224
data.flow.dest_ip	2a00:1450:4019:0003:0000:0000:0000:0006
data.flow.dest_port	443
data.flow.pkts_toclient	15
data.flow.pkts_toserver	28
data.flow.src_ip	2404:3100:1457:f606:e62b:5cba:489e:338a
data.flow.src_port	57858
data.flow.start	2025-06-14T13:44:40.796542+0500
data.flow_id	43423212754039.000000

